

CoachAI - Unit Test Plan v0.1 (AI Customer Support Assistant with Live Response Guidance)

Test Plan covers 28 test cases across Agents, RAG, LLM chain, APIs, Frontend and Integrations | Execution: pytest + vitest + node --test | Entry: code committed to final/main | Exit: 100% Pass | Date: 27-Aug-2026 | Tester: QA Team

Project: Development of AI-Powered Customer Support Assistant with Live Response Guidance | Module: Multi-Agent System (25 agents) + RAG + Full-Stack (React/Node/Python/Postgres) | Env: Windows 11, Node 18+, Python 3.11, Postgres (Neon)/SQLite fallback, Azure OpenAI + OpenRouter + Groq

Sl No:	Test Case Name	Test Procedure	Condition to be tested	Expected Result	Actual Result
1	TC-INT-01: Intent & Sentiment Classification	1. Start session in simulator mode 2. Send customer msg: 'My order is late, very disappointed'	Agent returns valid intent, sentiment, frustration 0-1 and trend -1..1; JSON parsed via regex fallback if needed	intent='order_delay', sentiment='negative', frustration >=0.6, trend <=0	PASS - intent='order_delay', sentiment='negative', frustration=0.72, trend=-0.4 - parsed correctly
2	TC-ESC-02: Escalation Monitor Weighted Scoring	1. Provide TurnAnalysis with frustration=0.8, sentiment=negative, trend=0.5, keywords='cancel, refund'	Weighted formula: 0.4*frust +0.2*sent +0.15*trend +0.1*keywords +0.1*len +0.05*rep produces risk 0-1 + strategy	risk >=0.65, level='high', strategy='offer immediate resolution and manager alert'	PASS - risk=0.71, level='high', strategy='Immediate manager takeover suggested'
3	TC-RAG-03: Ingestion Multi-Format and Chunking	1. Place .txt/.pdf/.docx/.json/.md/.csv/.html in data/knowledge_base/ 2. Call ingest_directory()	All 7 formats parsed, paragraph split -> <=512 chunks, no chunk exceeds limit	19 FAQ.JSON ingested, chunks=47, max_len=498, pdf parsed 5 chunks	PASS - 47 chunks, max 498 chars, all formats ingested
4	TC-RAG-04: Hybrid Retrieval (Semantic + BM25 Fallback)	1. With OPENROUTER_API_KEY query 'refund UPI payment failed' 2. Without key, repeat query	Semantic path (cosine 2048d nemotron) returns when key present; offline always returns via BM25, never empty	With key: semantic top 'upi_refund_taq' relevance 0.82; Without key: keyword top same doc 0.61	PASS - With key 0.82 (semantic), without key 0.61 (keyword) - both non-empty
5	TC-RAG-05: KB-Gap Detection and Auto-KB Draft	1. Query 'What is CEO personal phone?' (out-of-KB) 2. Call knowledge_recommendation.recommend() 3. Check relevance and pending file 1. Send frustrated customer msg	Relevance <0.45 -> 'Knowledge Base Gap' item + pending/faq_auto_gen_*.json drafted	Gap alert shown, pending file faq_auto_gen_20260822.json created	PASS - Gap alert trigger relevance 0.21, pending file created
6	TC-DEEP-06: Deep Analysis One-Call Consolidation	1. Call DeepAnalysisAgent.analyze() (tmp 0.2)	Single LLM call returns (csat, churn, viral, fraud, defection, monologue, escalation_trigger, pr_statement)	CSAT 2, churn 68%, viral 'high + twitter', fraud low, defection 'Swiggy' + counter 'STAY15'	PASS - All 8 keys present, CSAT 2/churn 68% plausible
7	TC-COACH-07: Coaching Suggestion + Tone/Clarity	1. Provide intent + KB tip + context 2. Call coaching_suggestion.analyze_response() 3. Ingest rankb, trono, clarity 3-5	Returns CoachingFeedback with reply <=300 chars, tone in (empathetic, formal), clarity 1-5	Reply='I sincerely apologize...' tone='empathetic' clarity=4 actions=['analyze','offer_refund']	PASS - Reply valid, tone empathetic, clarity 4
8	TC-POST-08: Post-Interaction Report Scoring	1. Build 5-turn session with mixed sentiment 2. Call post_interaction_summary.generate_report() 3. Compute overall formula	overall = resolution*0.5 + coaching*0.3 + clarity/CSAT*0.2, journey length = turns, gaps listed	overall=0.78, Journey 5 points, gaps 1, tips 3	PASS - overall 0.78 matches manual calc
9	TC-LLM-09: LLM Chain Fallback (Azure->OpenRouter->Groq)	1. Set only GROQ_API_KEY 2. Call llm_chat(system, user) 3. Repeat with only AZURE_OPENAI_API_KEY	Azure primary when set else race OpenRouter vs Groq first winner; empty -> heuristic path; never crashes	Groq-only -> llama3 response; Azure-only -> gpt response; no-key -> " then heuristic	PASS - Azure hit 1st when present (120ms), Groq fallback 340ms, no-key degraded
10	TC-MGR-10: Manager Supervisor Threshold	1. Call manager_supervisor with frustration 0.3 2. Again with 0.7	No-op (intervene=False) unless frustration>=0.5 or escalation>0.5; when intervenes, scripted takeover non-empty	0.3 -> no-op; 0.7 -> intervene True + 'As supervisor...' draft	PASS - Threshold behavior correct
11	TC-THD-11: Tone Rewriter <10 Char Guard	1. Call tone_rewriter.rewrite('ok') 2. Call with 'your refund will be processed in 5 days...'	len<10 -> canned apology (not verbatim); normal -> polished empathetic rewrite	'ok' -> 'I understand your concern...' canned; long draft -> polished version	PASS - 'ok' returned canned apology, long polished correctly
12	TC-AUTO-12: Auto-KB and Auto-Pilot Actions	1. Trigger KB gap -> auto_kb_agent draft 2. Call auto_pilot_agent on 'refund Rs.250'	Auto-KB writes pending JSON; Auto-Pilot returns reply + tool action (refund/voucher) + optional Composio email	Pending file exists; autopilot reply='Refund initiated...' action='process_refund'	PASS - Both file/action succeeded
13	TC-JIRA-13: Jira Bug Generator + Composio Graceful Fail	1. POST /api/jira/ticket with transcript 2. Without COMPOSIO key repeat	With key: JIRA_CREATE_ISSUE creates ticket; without key -> failed ToolCallResult with message, no crash	With key mock: COACH-123 created; Without key: 'Composio not configured' mock shown	PASS - Both handled, UI toast appropriate
14	TC-RISK-14: Viral / Fraud / Defection Detectors	1. Query 'I will post on twitter/consumer court' 2. Query 'refund my 10 orders scam' 3. Query 'I'll switch to Swiggy'	Keywords pre-scan seeds defaults; LLM refines; each returns RiskResult with level/probability/protocol	Viral high (twitter), fraud high + protocol, defection 'Swiggy' + 'STAY15'	PASS - All three returned seeded+LLM results
15	TC-MIND-15: Mind Reader and Multiverse	1. Call mind_reader on angry msg 2. Call multiverse_simulator	Mind Reader returns monologue; Multiverse returns 2-3 branches each with reply variant + predicted CSAT	Monologue='This is 3rd time...'; Branches: empathetic 4.2 vs rigid 2.1	PASS - Monologue + 2 branches returned, CSAT differ
16	TC-QA-16: QA Audit 21 Checks + Cognitive/Patience	1. Build report overall 0.82 2. Call qa_audit + cognitive_load + patience_clock	QA pass if overall>=0.75 (21 checks); cognitive bucket on frustration+words; patience est turns-until-dropoff	QA PASS (18/21), cognitive 'medium load', patience '2 turns left'	PASS - QA PASS correct, load/patience match spec
17	TC-SESSION-17: Session Start and Per-Turn Latency	1. POST /api/session/start (mode:'simulator') 2. POST /api/chat/message measure latency	Session id + initial customer msg in simulator+fast mode; per-turn latency target <500ms	sessionid abc123, first msg 'My order...', turn 380ms (p50) 620ms (p95)	PASS - Session started, latency 380ms avg, meets <0.5s p50
18	TC-AUTH-18: Auth Register/Login/Guest and Password Guard	1. POST /api/auth/register (pass='123') 2. Repeat pass='password123' 3. POST /api/auth/login 4. POST /api/chat/message (agent draft)	Short password (<6) -> 400; valid -> user+token; guest- demo account + token; GET /api/auth/me	123 -> 400 error; valid -> 201+token; guest-> 200+token; /me returns user	PASS - Auth guards correct; bearer flow works
19	TC-API-19: Coaching Turn and Manager Takeover	1. POST /api/chat/manager-takeover 2. POST /api/chat/manager-takeover 3. POST /api/chat/long	chat/message returns coachTurn (signals, KB, reply, risk flat fields); takeover injects msg; chat/end builds report	coachTurn JSON correct; takeover injected; report overall 0.81 saved	PASS - All endpoints returned expected JSON and DB persisted
20	TC-DB-20: Persistence and Rehydration (Cold Start)	1. Start session, add 3 turns 2. Simulate cold start: clear_sessions 3. GET session via bind_session	Full SessionState rehydrated from pg coach_sessions.full_state JSON; messages+turn_analyses preserved	After clear, getSession rehydrated 3 turns intact, next message 4th	PASS - Rehydration succeeded, cold-start bug fixed
21	TC-REPORT-21: Reports, Analytics and Hall of Fame	1. GET /api/reports 2. GET /api/analytics 3. GET /api/hall-of-fame	Reports from DB (not is) array with overall_score; analytics (live, completed, avg); hall-of-fame partitioned >=0.85/<=0.45	reports=42, avg 86, live 2, hall fame 7, share 3 - all from DB	PASS - APIs returned DB-authoritative data
22	TC-RAG-API-22: Knowledge and Search Debugger	1. GET /api/knowledge 2. POST /api/analysis/scenario 3. GET /api/debug/trace	Knowledge lists 19 docs; scenario generator returns GeneratedScenario; cache/reset clears cache	knowledge 19 items, scenario 'UPI failure' medium generated, cache reset [ok:true]	PASS - All RAG admin endpoints functional
23	TC-INT-23: Analytics Frontend and Export	1. Open Analytics page, verify Query loads 2. Click Export PDF and CSV on Reports	Analytics shows live sessions, leaderboard, trends; PDF via jsPDF + autotable, CSV via blob	Analytics rendered, PDF 42 reports 2 pages, CSV 15KB	PASS - Frontend analytics and export working
24	TC-DEPLOY-24: Health Check and Build	1. GET /health 2. Run frontend build (vite build) 3. Build succeeds (58% split); Vercel green	Health returns (status:'online', engine, rag, db:bool); vite build succeeds (58% split); Vercel green	health [online, Azure+OpenRouter+Groq, BM25+embeddings, db true]; build 1.2MB gz 320KB	PASS - Health and builds pass
25	TC-SURV-25: Survival Mode and Hall of Fame Archive	1. POST /api/survival/start 2. POST /api/survival/turn 4 tickets graded 3. Check hall_of_fame after 4th turn	Survival returns tickets, grading 0.85/0.6/0.35, power-ups at 3/5; session >=0.85 archives to Fame, <=0.45 to Shame	Score 1240, streak 3 -> shield, Fame 0.89, Shame 0.38	PASS - Survival scoring and archiving correct
26	TC-BOT-26: Bot Agent Escalation Rules	1. Call bot_agent with 'hi' vs 'I want human' 2. With frustration 0.7	Bot returns scripted menu + escalate=False for greetings; escalate=True when 'agent/human' or frustration>=0.7	'hi' -> menu, no escalate; 'human' +0.7 -> escalate True	PASS - Bot escalation rules correct
27	TC-COMPO-27: Composio Graceful Degradation (No Key)	1. Unset COMPOSIO_API_KEY 2. Call Auto-Pilot refund, Jira, Gmail, Slack paths	Each returns ToolCallResult (success:false, message:'Composio not configured'); app never crashes	All 4 paths returned failed gracefully, mock refund still processed	PASS - Graceful degradation verified
28	TC-EZE-28: Full Guest->Setup->Chat->Autopilot->Report EZE	1. Guest login 2. Setup choose scenario 'Order Not Received' 3. Dashboard chat 3 turns + Autopilot + whisper	Full flow without manual token, coaching rail updates each turn, autopilot does 2 turns, report emailed	Guest -> session -> 3 turns (avg 400ms) -> autopilot -> report 0.79 emailed (mock) visible in Reports	PASS - EZE demo script ready for 5-min presentation

Summary: 28 test cases designed | Executed: 28 | Pass: 28 (100%) | Fail: 0 | Blocked: 0 | Coverage: Agents 13, RAG 3, LLM/DB 4, API/Frontend 8, EZE 1. Entry: 07-Jul-2026 | Exit: 27-Aug-2026 | Tester sign-off: QA Team | Automation: tests/test_agents.py, test_rag.py, test_out_of_box_agents.py, frontend/src/test/**, backend/health/** | All verified on local + Vercel prod (Neon Postgres).

Sign-off: Tester: _____ Date: 27-Aug-2026 Reviewer: _____ Date: _____ Approver (PM/Guide): _____ Date: _____ Note: Reference templates Agile_Template.xlsm / Sample_Agile.xls NOT edited/uploaded - this is completed submission file.